

KOTA NAGA ABHINAY

Ongole, Andhra Pradesh | +91 9391127263 | abhinaykota2312@gmail.com

 LinkedIn |  GitHub

Experience

Research Intern – Crowd Monitoring and Safety Analytics

Jun 2025 – Aug 2025

SRM AP University

Under Prof. (Dr.) Niraj Upadhyaya

- Reviewed and synthesized **20+ research papers** on AI-based crowd monitoring and safety analytics, comparing methodologies and findings.
- Evaluated **UAV- and RFID-based monitoring approaches** across latency, reliability, scalability, and detection accuracy.
- Analyzed approaches for **crowd density estimation and abnormal activity detection** in intelligent monitoring systems.
- Prepared technical documentation and contributed findings to research discussions and system evaluation.

Projects

Synexa – Advanced RAG and Document AI Platform

 GitHub

Python, FastAPI, FAISS, BM25, MongoDB

- Built an end-to-end **RAG platform** for PDF and DOCX document intelligence with hierarchical chunking, document citations, and multi-provider LLM support.
- Designed hybrid dense and sparse retrieval using **FAISS and BM25** with Reciprocal Rank Fusion (RRF), improving reported context recall by approximately **35%**.
- Implemented two-stage retrieval with **cross-encoder reranking**, raising reported Context Precision@5 from **0.62 to 0.89**.
- Implemented **parent-child chunking, HyDE query expansion, and parent-context reconstruction** to improve retrieval quality for complex queries.
- Built quantitative RAG evaluation for **context precision, context recall, faithfulness, and latency**; SSE streaming reports **less than 200 ms TTFT**.

Payment Performance and Explainable Risk Analytics

 GitHub

Python, Pandas, Scikit-learn, XGBoost, SHAP, DuckDB, FastAPI

- Built an end-to-end payment risk platform processing **500K+ transaction records** with validation, feature engineering, ML scoring, and real-time serving.
- Developed an **Isolation Forest anomaly detection** pipeline with SHAP-based explainability and actionable risk reason codes.
- Implemented an **XGBoost risk classification** workflow with behavioral features for amount spikes, velocity bursts, device and location changes, and payment patterns.
- Built a **FastAPI risk-scoring microservice** and Streamlit dashboard; the project reports sub-**30 ms P99** execution for real-time evaluations.
- Used **DuckDB, dbt, and Pandera** for analytical modeling, transformation, schema validation, and reliable data processing.

Technical Skills

Languages: Python, C++, SQL

Machine Learning: Scikit-learn, XGBoost, Isolation Forest, SHAP, NLP, Feature Engineering, Machine Learning

LLM and Retrieval: RAG, FAISS, BM25, Reciprocal Rank Fusion, Cross-Encoder Reranking, Sentence Transformers, Google Gemini API

Data and Analytics: Pandas, NumPy, Polars, DuckDB, dbt, Pandera, EDA

Backend and Databases: FastAPI, MongoDB, MySQL, REST APIs

Tools: Git, GitHub, Jupyter Notebook, VS Code

Education

SRM AP University

2023 – Present

B.Tech in Computer Science and Engineering (AI and ML)

CGPA: 8.92/10

Expected Graduation: 2027

Narayana Junior College

2021 – 2023

Intermediate (MPC)

95.9%

Certifications

- MongoDB Associate Developer Certification
- EDA and Data Visualization in Data Science – Scaler Topics